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Kelp-bed dynamics across scales: Enhancing mapping capability with remote sensing and GIS

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ABSTRACT

Kelp are important drivers of productivity and biodiversity patterns in cold-water and nutrient-rich rocky reefs. Scuba- and boat-based methods are routinely used to study submerged kelp beds. However, these time-consuming and labor-intensive methods enable monitoring of beds or the factors and processes that control their distribution over only small spatial (few 100s of m^2) and temporal (< 5 years) scales. Remote sensing and geographic information system (GIS) technologies are increasingly used to compare marine species distribution over multiple spatiotemporal scales. However, there is currently no clear framework and limited demonstration of their potential for studies of broad-scale changes in completely submerged kelp beds. The present study aims to establish the foundation of a simple, accessible, and robust set of remote sensing and GIS-based methods to address key questions about the stability of subtidal kelp beds across multiple spatial and temporal scales. It tests the suitability of conventional image classification methods for mapping kelp from digital aerial (acquired on board a helicopter) and satellite (SPOT 7) imagery of ~250 ha of seabed around four islands in the Mingan Archipelago (northern Gulf of St. Lawrence, Canada). Three classification methods are compared: 1) a software-led unsupervised classification in which pixels are grouped into clusters based on similarity in spectral signature among pixels; 2) a software-led supervised classification in which pixels are assigned to categories based on similarity in the spectral signature of pixels and that of reference data from each category; and 3) a visual classification carried out by a trained observer. Supervised classification of satellite imagery and visual classification of aerial imagery were the top methods to map kelp, with overall accuracies of 89% and 90%, respectively. Unsupervised classification of both types of imagery showed poor discrimination between kelp and non-kelp benthic classes. Kelp bed edges were more difficult to identify on satellite than aerial imagery because the former presented poorer contrasts and a lower spatial resolution. Kelp bed edges identified with visual classification appeared artificially jagged for both types of imagery, mainly because of the coarse (225-m^2) spatial units used for this classification. Kelp bed edges were smoother on maps created with the unsupervised and supervised classifications, which used 1-m-pixel images. The present study demonstrates that conventional remote sensing and GIS methods can accurately map submerged kelp beds over large spatial domains in the Mingan Archipelago or in other benthic systems with similar oceanic conditions and a largely dichotomous (kelp-barrens) biological makeup.

1. Introduction

Kelp (large brown seaweeds of the order Laminariales) are key drivers of productivity and biodiversity patterns in cold-water and nutrient-rich, shallow rocky reefs (Dayton, 1985; Tegner and Dayton, 2000). Kelp typically form structurally complex aggregates, known as kelp beds or forests, which provide critical habitat to a variety of fish and invertebrates (Estes et al., 2004; Ling, 2008; Steneck et al., 2002). Worldwide, large-scale shifts from kelp-dominated to urchin-dominated community states have occurred following increases in the intensity of urchin grazing on kelp, or as a result of climate-driven shifts in species distribution (Ling et al., 2015; Steneck et al., 2002; Vásquez et al., 2007; Wernberg et al., 2016). Although these shifts between community states affect 10s to 100s of km^2 of coastal habitats (Filbee-Dexter and Scheibling, 2014; Krumhansl et al., 2016; Moy and Christie, 2012), most studies of subtidal kelp systems, including distributional aspects, have been conducted over small spatial (few hundreds of meters) and temporal (less than five years) scales mainly because of the limitations of the time-consuming and labor-intensive scuba techniques typically employed (Gagnon et al., 2004; Lauzon-Guay and Scheibling, 2007;

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Scheibling et al., 1999; Van Rein et al., 2009). Such studies provide valuable information about local kelp bed dynamics. However, distributional patterns of kelp and the underlying ecological drivers may vary across scale (Cavanaugh et al., 2010; Schneider, 2001; Turner et al., 1989; Van Rein et al., 2009), and hence repeated measurements of kelp abundance and the ecological drivers of change at broad spatial and temporal scales are desirable over inferences from small scale patterns.

In the northern Gulf of St. Lawrence (Canada), mixed kelp beds dominated by *Alaria esculenta* (Linneaus) and *Laminaria digitata* (Hudson) often form a fringe from the lower intertidal to depths of ~7 m. Dense (up to 400 individuals m^{-2}) aggregations of green sea urchins, *Strongylocentrotus droebachiensis* (Müeller), known as urchin fronts, typically form at the lower edge of the beds, more so in summer. Urchin fronts destructively graze the beds as they move in shallower water, leaving behind extensive barrens largely devoid of fleshy macroalgae (Gagnon et al., 2003; Gagnon et al., 2004; Himmelman, 1991). The Mingan Archipelago is a group of > 40 islands along the northern shore of the Gulf of St. Lawrence, between Anticosti Island and Québec's Middle North Shore. The shallow seascape around these islands presents the biological characteristics described above, with a spatial dominance of barrens over kelp beds within the first ~10 m of water (Gagnon et al., 2003; Gagnon et al., 2004; Himmelman, 1991).

Scuba-based research carried out around many islands since the early 1980s, suggests that urchin barrens can persist for decades, mainly because of sustained urchin grazing (see Gagnon et al., 2003, 2004, 2005; Himmelman, 1991; Narvaez Diaz, 2018). This situation differs from that in more southerly regions of eastern Canada with warmer waters, in particular Nova Scotia, where outbreaks of a pathogenic amoeba have sporadically decimated urchin populations, causing large-scale shifts in kelp/barrens community states (Filbee-Dexter and Scheibling, 2012, 2014; Lauzon-Guay et al., 2009). The presumed stability of the kelp bed and barrens states across the Mingan Archipelago is not backed by a rigorous, broad-scale assessment of temporal changes in the spatial extent of each community state. Also, factors that control the broad-scale stability, or lack thereof, of these community states are largely unknown. Synoptic approaches that depend less on scuba techniques, more on mass acquisition and analysis of data, must be developed to properly address the frequency, causes, and consequences of community phase shifts in this and other systems, at scales that enable significant gains in ecological knowledge and ability to manage marine resources.

Broad-scale marine habitat mapping programs increasingly rely on remote sensing and geographic information system (GIS) technologies (Finkl and Makowski, 2014; Green et al., 2000; McCarthy et al., 2017), which provide several key advantages over traditional scuba-based techniques. First, acquisition of imagery (or other types of data) with airborne or spaceborne sensors can be much quicker and cover much larger areas, including in remote locations (Cavanaugh et al., 2010; Deysher, 1993; Green et al., 2000; Van Rein et al., 2009). Orbiting satellites revisit the same locations on the earth's surface at regular intervals, thus acquiring core information for community change analysis (Brooks et al., 2015; Lyons et al., 2011; Yang and Yang, 2009). High-resolution satellite images covering a variety of spatial domains are publicly (and sometimes freely) available (e.g., Worldview-1 to 4 [0.31 to 0.46-m pixels], Quickbird [0.65-m pixels], SPOT7 [1.5-m pixels], PlanetScope [3-m pixels], and RapidEye [5-m pixels]), therefore facilitating access to core data. Secondly, images can be georeferenced, projected, and analyzed in a GIS, and spatially matched with environmental factors to explore the causes and consequences of changes of a given community component (Garza-Pérez et al., 2004; Kendrick et al., 2000; Lathrop et al., 2001; Silva et al., 2008). Despite these advantages, mapping of subtidal benthic communities based on optical imagery acquired from above the sea surface can be challenging, mainly because of the exponential attenuation of sunlight passing through a water column (Green et al., 2000; Jerlov, 1976). This phenomenon, which further varies with the type of oceanic water (Jerlov, 1976; Maritorena et al., 1994; Mobley, 1994) ultimately alters the spectral values of the seabed and benthos recorded by the camera and, in turn, the ability to correctly classify benthic community types (Ackleson, 2003; Gagnon et al., 2008; Green et al., 2000). Because of this challenge, studies of submerged vegetation from remotely acquired optical imagery have largely been in benthic habitats overtopped by optically shallow (low-turbidity) waters (Andréfouët et al., 2004; Gullström et al., 2006; Hedley et al., 2012; Hochberg and Atkinson, 2003; Hu et al., 2015; Kendrick et al., 2000; Lyons et al., 2011).

Most studies that have used optical remote sensing to map kelp have focused on tall species (e.g. the giant kelp, *Macrocystis pyrifera*) that form forests and grow tissues at or near the sea surface, therefore substantially reducing the risk of misclassification (Bell et al., 2015a,b; Cavanaugh et al., 2010, 2011; Friedlander et al., 2018; Grove et al., 2002; Nijland et al., 2019; Stekoll et al., 2007). The greater challenge of detecting largely submerged, low-lying kelp species from above the sea surface, like is the case in the Mingan Archipelago, might partly explain the rarity of studies that have mapped completely submerged kelp beds with optical remote sensing (Casal et al., 2011; Malthus and Karpouzli, 2003; Simms and Dubois, 2001; Vahtmäe et al., 2012). Water column correction techniques can be applied to improve the accuracy of classification maps of benthic habitats (Hoang et al., 2015; Sagawa et al., 2012; Zoffoli et al., 2014), yet their complexity may limit their implementation. Remote sensing and GIS technologies evolve rapidly (Finkl and Makowski, 2014; Frohn and Lopez, 2017; McCarthy et al., 2017), becoming increasingly powerful and complex. As highlighted by Andréfouët (2008), the perspective and objectives of "map producers" (e.g. remote sensing specialists) whose main goal is to develop highly effective methods, differ from those of "map users" (e.g. scientists and managers) who are primarily interested in creating trustable and adaptable products (e.g. habitat classification maps and accuracy metrics) in simple ways. As map users, those with a prime interest in studying submerged kelp bed dynamics, i.e., marine benthic ecologists, may not be familiar with, or have access to, specialized, cutting-edge remote sensing technologies. These non-specialists would therefore benefit from a demonstration of the gain in information that simple and accessible remote sensing and GIS technologies can provide.

The present study is the first of a series aimed at developing and presenting a simple and accessible, yet robust set of remote sensing and GIS-based approaches to address key questions about the stability of kelp beds across multiple spatial and temporal scales. The series uses the Mingan Archipelago as a study system because of its relative simplicity (see details in section 2.1) and the opportunity that previous scuba-based studies offer to compare patterns and drivers of kelp stability across scales (Gagnon et al., 2003, 2004, 2005; Himmelman and Dutil, 1991; Narvaez Diaz, 2018). The present study's main objective is to identify the most reliable methodology to quantify the distribution of completely submerged kelp that subsequent studies can use to generate core information to quantify biological and ecological patterns and their natural drivers. This objective is accomplished by testing the suitability of three conventional classification methods for kelp detection and mapping: (1) a software-led unsupervised classification in which clusters of pixels are identified based on the similarity in spectral signature among pixels and the user subsequently assigns the clusters to a cover class [in the present case, either "kelp" or "non-kelp"]; (2) a software-led supervised classification in which the user provides field and other reference data to identify the spectral characteristics of each cover class, then each pixel is assigned to a class by the software based on spectral similarity; and (3) a visual classification in which a trained observer identifies the presence of either cover class on the imagery across a grid of points. Each classification method was applied to digital aerial (acquired on board a helicopter) and satellite (SPOT 7) imagery covering nearly 250 ha of seabed around four islands. The imagery was acquired at times of optimal sea-state conditions to facilitate image classification. Each method's operational and logistical advantages and limitations are discussed to guide the choice of a particular methodology for systems similar to the Mingan Archipelago's but where access to resources may differ.

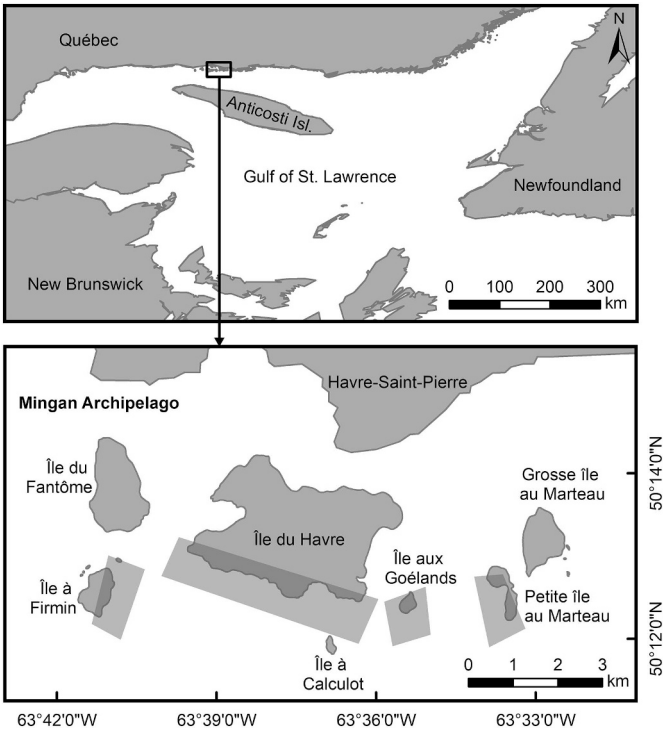

Fig. 1. Location of study areas (shaded polygons) at Île à Firmin, Île du Havre, Île aux Goélands, and Petite île au Marteau in the western sector of the Mingan Archipelago, northern Gulf of St. Laurence, eastern Canada.

2. Materials and methods

2.1. Study system

The study system encompassed the seabed around four islands in the Mingan Archipelago: Île à Firmin (westernmost), Île du Havre, Île aux Goélands, and Petite île au Marteau (easternmost) (Fig. 1). The seabed in this system is mainly composed of bedrock and boulders, with occasional patches of cobble, gravel, or sand in areas with low hydrodynamic forces (Gagnon et al., 2004). This system was chosen because it presents characteristics facilitating the application and validation of the methods described below. First, kelp beds composed mainly of *Alaria esculenta* and *Laminaria digitata*, and to a lesser extent *Agarum clathratum* (Dumortier), *Saccharina longicruris* (Bachelot de la Pylaie), and *Saccorhiza dermatodea* (Bachelot de la Pylaie), develop at depths of 0 to 7 m and are generally more abundant on southern, wave-exposed sides of the islands (Gagnon et al., 2004). These thick, dark-colored beds are followed in deeper water by green sea urchin (*Strongylocentrotu droebachiensis*) barrens colonized by a thin, light-colored layer of live and dead encrusting coralline algae, mainly *Lithothamnion glaciale* (Kjellman) (Fig. 2.B, Gagnon et al., 2004). Such a dichotomous division of the benthos into two biologically and spectrally distinct community types, hereafter termed "kelp" and "non-kelp", is compatible with the application of a simple image classification method. Second, the transition between kelp beds and barrens is often clear-cut, mainly because of grazing by urchin fronts advancing over the lower edge of the beds (Fig. 2, Gagnon et al., 2004). Sharp transitions between these two spectrally different community types further facilitate identification on imagery (Fig. 2.D and 2.E). Third, seabed slope is gentle, on average ~3% as calculated from bathymetry data, with no major sudden shifts in topography. Such a mild, gradual change in slope and the shallow depth range (0 to 7 m) on which the present study focuses limit effects of attenuation of light by the water column on benthic spectral signatures (Green et al., 1996, 2000), therefore eliminating the need to apply water column correction. All these characteristics reduce the likelihood of misclassifying both community types.

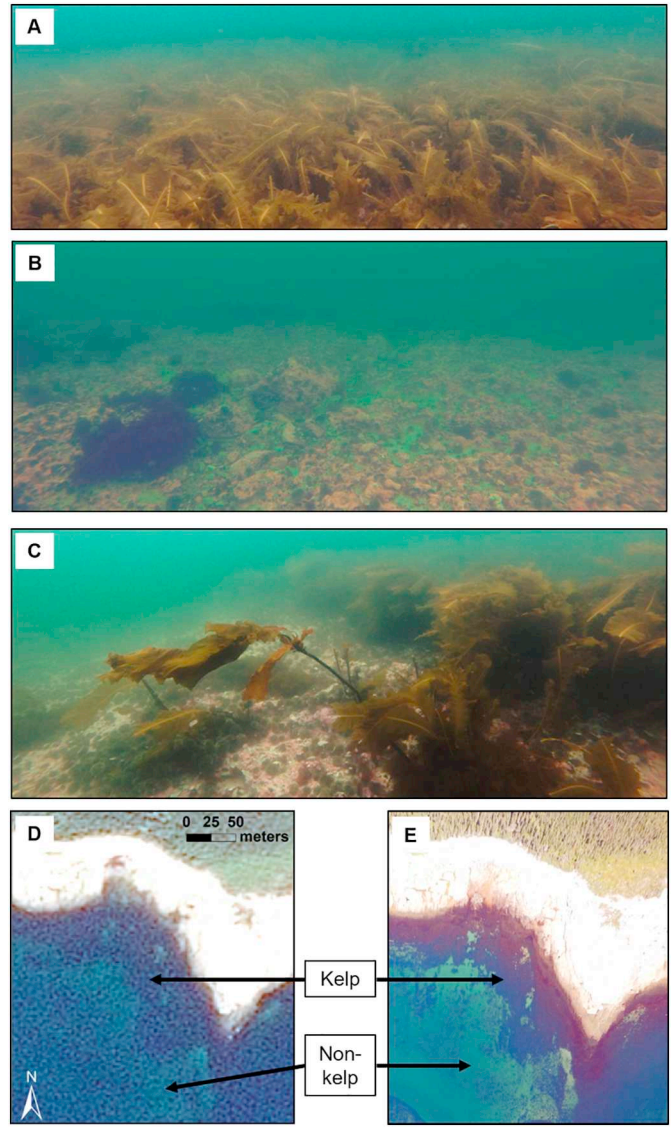

Fig. 2. A) Kelp bed [mainly *Alaria esculenta*] at a depth of ~2 m within the Île du Havre study area (photo: Anne P. St-Pierre). B) Green sea urchin (*Strongylocentrotus droebachiensis*) barrens with a few kelp patches at a depth of 6 m within the Île du Havre study area (photo: Anne P. St-Pierre). C) Sharp transition between the lower edge of a kelp bed and upper edge of an urchin barrens caused by an urchin grazing front advancing over kelp at a depth of ~2 m within the Île du Havre study area (photo: Anne P. St-Pierre). D) Satellite [SPOT 7] image of a portion of the Île du Havre study area. E) Aerial image of the same portion shown in D (photo: Patrick Gagnon).

Monospecific stands of the grazing-resistant, perennial kelp *A. clathratum* and annual brown seaweed *Desmarestia viridis* (Müller) that cover up to a few 10s of m^2 can develop in the barrens at depths of up to 15 m (Gagnon et al., 2003, 2004, 2005). *Agarum clathratum* stands are considered a stable component of barrens, whereas stands of *D. viridis* are ephemeral, with annual recruitment and die-off in late winter and early fall, respectively (Blain and Gagnon, 2014; Gagnon et al., 2003, 2004, 2005). These stands typically develop in areas of low to moderate wave action, along the lower edge of *A. esculenta* beds (*A. clathratum*) or in the barrens (*A. clathratum* and *D. viridis*) (Gagnon et al., 2003, 2004, 2005). While *A. clathratum* and *D. viridis* are not consumed by sea urchins and grow outside of mixed kelp beds in this region, they are not generally considered as part of kelp beds. However, these seaweeds could not be distinguished from mixed kelp beds on the imagery. As discussed later (see Discussion), there was low potential for

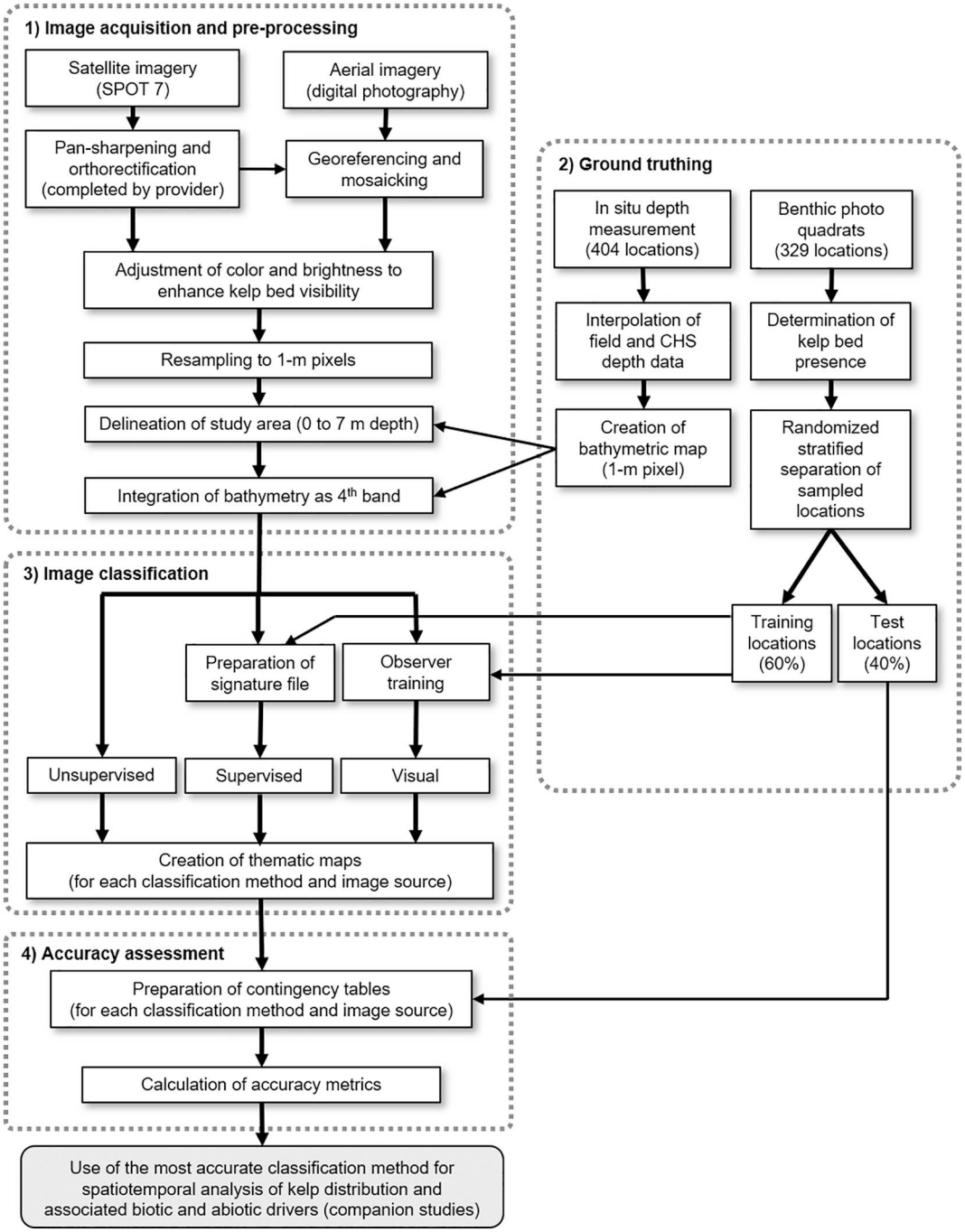

Fig. 3. Break down of steps in each of the four methodological modules used in the present study. (1) *Image acquisition and pre-processing*: satellite imagery of the study area was used as a base map to georectify [arrow within box 1] corresponding aerial imagery. Both sets of images were digitally balanced to enhance contrast between kelp and other benthic components. Bathymetric mask was applied to restrict analysis to shallow (0 to 7 m deep) seabed. (2) *Ground truthing*: bathymetric data acquired in situ with an echosounder and from the Canadian Hydrographic Service [CHS] were used to support delineation of each study area and image classification [arrows from box 2 to box 1]. Benthic photo quadrats acquired within each study area were used to validate the presence and absence of kelp on satellite and aerial imagery and support image classification and accuracy assessment [arrows from box 2 to boxes 3 and 4]. (3) *Image classification*: pre-processed satellite and aerial imagery and ground truth data were used to create maps of kelp distribution from three different classification methods: unsupervised, supervised, and visual. (4) *Accuracy assessment*: classification accuracy for each of the six thematic maps created was calculated based on test location ground truth data [arrow from box 2 to box 4].

misclassification of *D. viridis* stands as kelp and other non-kelp macroalgae were rare within the study area, covering only ~6% of the seabed (see Section 2.3). In total, 248.4 ha of seabed were classified, with respectively 15, 62, 13, and 10% of this surface at Île à Firmin, Île du Havre, Île aux Goélands, and Petite île au Marteau.

To identify the most accurate classification method (out of the three methods tested) for kelp detection and mapping in the Mingan Archipelago system, the procedures applied were grouped under four modules detailed in the sections below and summarized schematically (Fig. 3): (1) image acquisition and pre-processing [Section 2.2]; (2) ground truthing [Section 2.3]; (3) image classification [Section 2.4]; and (4) accuracy assessment [Section 2.5]. These modules combine satellite (*Satellite Pour l'Observation de la Terre*; SPOT 7), aerial (digital), and underwater (digital) seabed imagery of the study area, as well as

bathymetric data and conventional image classification techniques and accuracy assessment metrics. Some of the key remote sensing and GIS terms employed are **underlined and defined in a separate section (Appendix A)**.

2.2. Image acquisition and pre-processing

The two types of remotely sensed imagery used in the present study - one SPOT 7 satellite image and multiple aerial photographs acquired with a basic digital camera - align with the overarching goal of establishing simple and reliable ways to map subtidal kelp cover over large tracts of seabed. Original imagery differed in spatial coverage (175 to 1015 times larger on SPOT 7 than on aerial images), spatial resolution (14 to 320 times lower on the SPOT 7 image), and contrast (lower on the SPOT 7 image). These differences enabled the comparison of the accuracy of classification maps derived from a readily available commercial product (SPOT 7 image) and imagery that is logistically more challenging and time-consuming to acquire and process (aerial photographs). The SPOT 7 image, acquired on 11 August 2016, in the hour preceding low tide, encompassed the entire study system. It was taken after four days with low winds ($16.4 \pm 1.1\,\mathrm{km\,h^{-1}}$, based on data from Environment Canada [http://climate.weather.gc.ca]) and no precipitation and contained no cloud cover or surface waves. Aerial photographs were acquired sequentially with overlap from Petite île au Marteau to Île à Firmin on board a helicopter flown at an altitude of ~300 m. They were taken with a hand-held digital camera (Nikon Coolpix AW130) at low tide on 8 July 2016, after four days of low winds ($14.1 \pm 0.8\,\mathrm{km\,h^{-1}}$, based on data from Environment Canada [http://climate.weather.gc.ca]) and no precipitation. None of the images contained cloud cover or surface waves. We deliberately chose dates and times of image acquisition for each image source with best possible sea conditions (i.e. low waves, low winds, low turbidity) to facilitate image classification. These conditions minimized the amount of suspended particles in the water column and glare from sunlight reflection at the sea surface, enabling viewing the seabed clearly to depths of up to ~8 m on both satellite and aerial images (Fig. 2.D and 2.E). Water-column data enabling quantifying water turbidity were not available. Yet, water clarity was highest for the region at this time of the year (P. Gagnon, personal observations over multiple years). Logistical considerations prevented acquiring aerial photographs on a day for which a SPOT image was available.

One key aspect affecting the quality of image classification is the spectral resolution of images. In theory, a higher spectral resolution yields greater spectral differences among image pixels with different objects or biota, thus increasing separability (Green et al., 2000). Spectral resolution refers to the number and width of spectral bands in an image, such that images with multiple spectral bands (channels) generally have a greater spectral resolution and are preferred over panchromatic (one band) images (Green et al., 2000). Furthermore, in optically shallow waters like was the case in the present study, short (blue and green) wavelengths penetrate deeper than long (red and near-infrared) ones (Jerlov, 1976; Maritorena et al., 1994; Mobley, 1994). Consequently, sensors with a greater number of bands recording in the shorter wavelengths typically produce more useful imagery of completely submerged benthic features than sensors with a greater number of bands recording in the longer wavelengths (infrared wavelengths are completely filtered out within the first few centimeters of water; Lillesand et al., 2014). Yet, the expertise and software and hardware needed to process and classify images also increase with the number of spectral bands (Bioucas-Dias et al., 2013; Green et al., 2000). Accordingly, images with only few bands may provide sufficient resolution when dealing with low habitat complexity, like in the present study.

Both satellite and aerial images were true color composite images. The SPOT 7 image was pan sharpened to a spatial resolution (pixel size) of 1.5 m and orthorectified by the image provider (Land Info Worldwide Mapping LLC, Colorado, USA). The three bands forming the satellite image covered the 455 to 525 nm (blue), 530 to 590 nm (green), and 625 to 890 nm (red) ranges. Spatial resolution of aerial photographs varied between 7 and 20 cm depending on camera (helicopter) height and angle. Although the aerial photographs were 3-band (blue, green, and red) color composite images, the specific wavelengths recorded in each band is unknown. Aerial photographs were georectified and mosaicked (Lillesand et al., 2014) to create a single image of the four islands using the SPOT 7 image as the registration template and six to 14 photographs per island (based on the size of the study area and quality of the photographs) with side overlap among photographs. During mosaicking, 10 to 15 ground control points per photograph were used and a projective transformation was applied. This transformation was chosen over other common types of transformation because it yielded the lowest root mean square residual error (between 1.2 and 9.5 m), and the best visual match to the satellite imagery. During mosaicking, contrast and brightness of each aerial photograph were adjusted with ArcGIS' Contrast and Brightness function to facilitate kelp detection and reduce variation among photographs. Contrast and brightness were also adjusted on the SPOT 7 image to facilitate kelp detection. To facilitate data processing and analysis, all individual layers were converted to a common map coordinate system (Universal Transverse Mercator Projection, Zone 20 N, North American Datum 1983). All raster datasets were resampled to a 1-m grid cell size with the nearest neighbour resampling algorithm (Lillesand et al., 2014). Atmospheric and radiometric corrections, which correct for or calibrate irregularities in pixel values due to atmospheric distortion or instrumentation errors, were not applied because only one satellite image was used (Andréfouët, 2008; Green et al., 2017; Song et al., 2001). These corrections should be applied to calibrate pixel values prior to analysis when using multiple satellite images acquired at the same or different times (Green et al., 2017; Lillesand et al., 2014). The SPOT 7 image and mosaics of aerial photographs were cropped to keep only seabed between 0 and 7 m deep (see Sections 2.3). Light attenuation in the water column was indirectly accounted for by adding bathymetry as a 4th band to both types of imagery as a means of improving classification with the unsupervised and supervised classification methods (see Section 2.3).

2.3. Ground truthing

2.3.1. Bathymetry

Preliminary analysis of bathymetric charts for the Mingan Archipelago obtained from the Canadian Hydrographic Service (CHS) revealed inaccuracies in the 0 to 5 m depth range. Bathymetry across the study area was therefore acquired with a boat-mounted depth sounder (Lowrance Hook5 recreational fishfinder, vertical precision of 0.1 m) equipped with a GPS unit. A horizontal accuracy below 5 m was estimated for the GPS unit by comparing the position of features on the SPOT 7 image (e.g. docks, roads, rocky features across the study area) with *in situ* measurements of their position with the depth sounder's GPS unit. In total, 404 depth soundings were acquired, with 83, 226, 57, and 38 soundings at Île à Firmin, Île du Havre, Île aux Goélands, and Petite île au Marteau, respectively. Bathymetric data were corrected for tidal elevation at the time of measurement. This was done by subtracting tidal amplitude (precision of 0.1 m, based on water level observations by the Department of Fisheries and Oceans Canada [www.waterlevels.gc.ca]) from in situ depth measurements. A bathymetric chart encompassing the four study areas was created with ordinary kriging based on the self-acquired, corrected bathymetric data and the 10-m isobath on CHS' 2002 bathymetric chart. This chart, which had a grid cell size of 1 m (Fig. 4), was incorporated to the GIS platform used to classify images (see section 2.4), both to crop the SPOT 7 image and mosaic of aerial photographs (hereafter termed "aerial mosaic") to the targeted 0 to 7 m depth range, and as a means of potentially improving classification accuracy (Gagnon et al., 2008). The SPOT 7 image and the aerial mosaic contained three bands (layers) of spectral information for each image pixel; one for each of the red, green, and blue wavelengths. The addition of the 4th non-spectral bathymetric band (see above) created a 4-band raster.

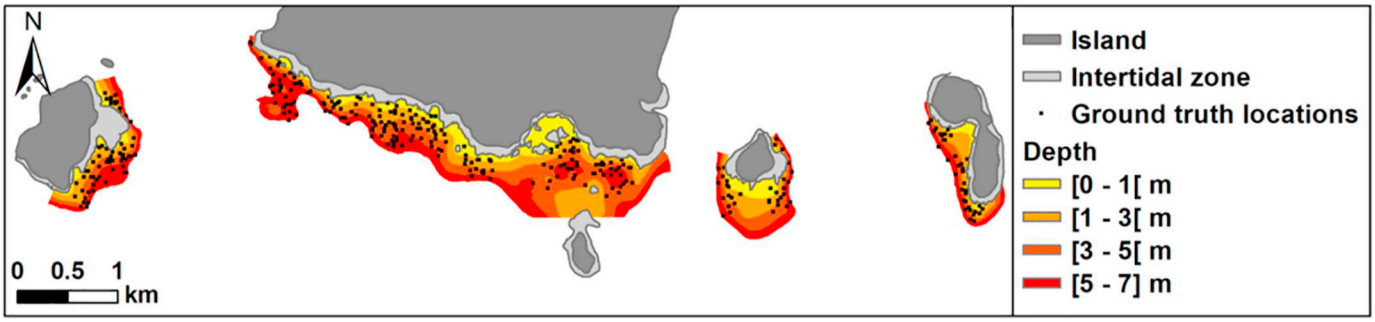

Fig. 4. Bathymetry of the study system used to map kelp from satellite and aerial imagery with locations of ground truth data acquisition (see summary of methodological details in Fig. 3).

2.3.2. Complementary data

The three image classification methods used required *in situ* data about the presence and absence of kelp for training purposes (in the case of supervised and visual classifications) and to calculate classification accuracies (see Sections 2.4 and 2.5). Kelp occurrence at multiple locations across the four study areas was determined between 23 July and 29 July 2016. To facilitate field work, sampling locations in each study area were pre-determined with a random point generator (Create Random Points tool in ArcGIS v10.3.1, Esri, 2015). The number of locations in each area was proportional to the size of the area, with a minimum distance of 15 m between locations. The nearly even distribution of the locations enabled a spatially balanced sampling design. However, 86 of the 415 pre-determined locations (~21%), were not sampled because they were deeper than the 7-m limit or too shallow to be safely accessed with the boat used for the survey. Of the 329 locations sampled 63, 197, 34, and 35 were at Île à Firmin, Île du Havre, Île aux Goélands, and Petite île au Marteau, respectively. At each location, a down-facing drop camera system (GoPro Hero 3 in an underwater housing, attached to a metal frame) was lowered to ~1 m above the seabed. The height of the camera allowed photographing the seabed within an 80 × 80-cm plot. In total, 231 locations contained one or more kelp sporophytes and 98 had none. The 329 locations were divided into: (1) a training dataset [60% of the locations] used to guide the supervised classification and to train an observer in recognizing kelp on the imagery prior to visual classification; and (2) a test dataset [40% of the locations] used to quantify the accuracy of each classification method [see Section 2.5]. To ensure that both datasets were spatially balanced among the four study areas and evenly represented each benthic cover class ("kelp" and "non-kelp"), a stratified, randomized selection method was applied to divide the locations into training and test datasets. This was done by randomly selecting (without replacement) and assigning to the training dataset 60% of the locations from each benthic cover class within the study area of each of the four islands and assigning the remaining 40% to the test dataset.

2.4. Image classification

Three image classification methods commonly used in benthic habitat mapping studies (Brown et al., 2011; Green et al., 2000) were considered: (1) unsupervised [software-led]; (2) supervised [software-led]; and (3) visual [observer-led]. Each method was applied to both the satellite imagery and aerial mosaic for comparison purposes. All methods employed a binary classification scheme, whereby the image was divided into either of two benthic classes: "kelp" or "non-kelp". Kelp species could not be discriminated given the type and resolution of the images.

Unsupervised classification was carried out in ArcGIS v10.3.1 with the Interactive Self-Organizing (ISO) Cluster Unsupervised Classification algorithm (Esri, 2015) and default settings (minimal number of pixel in a valid class = 20, sample interval = 10). This algorithm separates pixels of input raster images into a user-defined number of clusters based on similarities of pixels' spectral signatures. It is generally recommended that the number of clusters be at least five times that of cover (benthic) classes (two in the present study), and that each cluster contains at least 10 times as many pixels as the number of bands in the imagery (four in the present study; Green et al., 2000; Lillesand et al., 2014). For each type of imagery, outputs were created with various numbers of clusters (10, 15, 20, 30, and 40), with 20 providing the best spatial distribution of the two target classes based on visual comparisons with the imagery and complementary field data. Each cluster created by the ISO Cluster algorithm was subsequently assigned to either of the two benthic classes (kelp or non-kelp) by the user based on visual comparison with the original imagery and field data to create a thematic map of the spatial distribution of both benthic classes.

Supervised classification was carried out in ArcGIS v10.3.1 with the Maximum Likelihood Classification (MLC) tool (Esri, 2015). This technique consists of two steps. First, the training dataset created from ground truth data (see section 2.3) is used to identify, on the imagery, multiple areas representative of each benthic class. The digital signature of each area, which is based on the numerical values contained in the four bands [red, green, blue, and bathymetry] of each pixel that form the area, is used to create a signature file, which defines the mean numerical values (and their variance) for each benthic class, on each type of imagery. Second, the MLC tool assigns each pixel in the input raster image to a benthic class based on the similarity between the pixel's digital signature and the signature of the classes identified in the signature file. Once all pixels have been assigned to classes, the MLC tool produces a thematic map showing the distribution of the two benthic classes.

Visual classification was carried out by an observer (A. P. St-Pierre) trained to recognize kelp presence and absence on the imagery from the position and cover type of ground-truthed locations. A grid with evenly spaced points was overlaid on each type of images. The distance between adjacent points, 15 m, was consistent with the approximate scale of earlier studies of kelp bed dynamics in the Mingan Archipelago (Gagnon et al., 2004; Gagnon et al., 2005). The 225-m^2 (15 × 15 m) area surrounding each point was categorized as "kelp" when 50% or more of the area contained kelp or as "non-kelp" when kelp was < 50%. Visual classification is based on human perception of shapes, color, and contrasts within an image rather than on numerical values contained in the various bands of information of individual image pixels. Consequently, the observer must be trained to recognize the visual characteristics of kelp on the imagery prior to conducting the classification. Observer training in the present study was achieved by overlying the training dataset created from the ground truth data on the imagery to identify areas where the field data confirmed the presence or absence of kelp. The observer compared the color, brightness, tone, and shape of known kelp and non-kelp areas, and applied this knowledge to classify the rest of the imagery. In general, the dark (blue-brown) areas, covered in kelp, contrasted sharply with the surrounding light (greenish) areas, devoid of kelp (Fig. 2.D and 2.E). Thematic maps were prepared based on the classification of each point in the grid, for the satellite and aerial imagery separately.

2.5. Accuracy of classification

To identify which combination of classification method and image source yielded the most precise kelp distribution map, the accuracy of the six thematic maps produced with the unsupervised, supervised, and visual classification methods (applied to the satellite and aerial imagery separately) was quantified and compared with accuracy metrics commonly used in the field of remote sensing (Green et al., 2000; Lillesand et al., 2014). This was first done qualitatively by examining the thematic maps visually, to evaluate if patterns suggestive of artifacts from the original imagery (e.g. sharp differences in contrast between images) were present. The proportion of the study area classified into each benthic class was calculated for each thematic map and compared. The similarity among thematic maps was assessed by calculating the proportion of spatial overlap in the distribution of each benthic class, to identify the location and extent of areas consistently classified as kelp or non-kelp by all classification methods. The proportion of spatial overlap was also calculated for all possible pairs of thematic maps. To quantitatively assess classification accuracy, contingency tables, also known as confusion or error matrices, were constructed based on the test dataset (40% of the ground truth data). In contingency tables, the agreement between the true benthic class observed at a location *in situ* (based on analysis of ground truth data) and the benthic class attributed to the same location by the classification is assessed for each location sampled in the field and compiled for each benthic class individually. Hence, the proportion of correctly or incorrectly classified locations in a class can be compared with that of the other classes (see Appendix B for contingency tables from the present study). Overall accuracy and kappa coefficient were both calculated from each of the six contingency tables. Overall accuracy, which ranges from 0 to 100%, indicates the proportion of correctly classified locations (all classes included). Kappa coefficient, which ranges from -1 to 1, indicates how well the classification agrees with ground truth data, with values ≤ 0 indicating chance agreement (a meaningless classification) and values of 1 a perfect agreement between ground truth and classified data. Producer's accuracy and user's accuracy were used to compare the accuracy of detection of each benthic class separately. Producer's accuracy is the proportion of locations correctly classified in a given class, whereas user's accuracy is the proportion of locations classified in a given class that are actually present under the same class on the seabed. See Landis and Koch (1977) and Lillesand et al. (2014) for further discussion of these statistics.

3. Results

3.1. Visual assessment of thematic maps

The satellite imagery (which encompassed the entire study area) appeared grainy, with relatively coarse spatial resolution and poor contrast between benthic features (Fig. 5.A). Kelp and non-kelp pixels on the thematic map created from unsupervised classification were highly segregated, forming small, discontinuous clusters scattered across the seabed around the four islands (Fig. 5.B). With the supervised classification, kelp formed larger, more continuous patches, which also better matched their true distribution on the satellite image (Figs. 5.A and 5.C). In comparison, kelp cover appeared higher with the visual classification, with long and jagged boundaries (Figs. 5.A and 5.D).

The aerial imagery exhibited better contrasts and delineations between spectrally different benthic features than the satellite imagery (Figs. 5.A and 6.A). However, contrary to the satellite imagery, there was considerable variation in color, tone, and contrast from one photograph to the next (Fig. 6.A) despite efforts to spectrally equalize them during mosaicking. The consequences of this spectral variation were particularly apparent in the thematic map obtained from the unsupervised classification on which: (1) large tracts of seabed classified as kelp contained no kelp on the aerial imagery and vice-versa; and (2) delineations between areas with or without kelp sometimes followed the margins between adjacent photographs (Fig. 6.B). Kelp distribution with the supervised classification more closely matched that on the aerial imagery, although some artifacts similar to those noted for the unsupervised method remained, particularly around Île à Firmin and along the eastern half of Île du Havre (Fig. 6.C). As observed with the satellite imagery, kelp cover appeared higher with the visual classification, with coarse boundaries encompassing smaller areas with no kelp on the aerial imagery (Figs. 5.D, 6.A, and 6.D).

3.2. Spatial overlap in distribution of benthic classes

Overall, the six classification methods similarly classified 32% of the study area as kelp (25%) or non-kelp (7%) (Table 1). The remaining 68% was classified differently by at least one classification method and was scattered across the area, particularly at the deeper end (5–7 m) with 86% of the seabed classified differently (Figs. 4 and 7, Table 1). The proportion of seabed classified as kelp varied among classification methods, from 52% in the unsupervised classification of aerial imagery to 74% in the visual classification of satellite imagery (Table 2). With a spatial overlap of 81% in seabed classified as kelp, supervised classification of satellite imagery and visual classification of aerial imagery yielded the most similar kelp bed distributions, followed closely by visual classifications of both types of imagery at 80% overlap in kelp distribution (Table 2). As to seabed classified as non-kelp, supervised classification applied to both types of imagery yielded the most congruent results with 80% overlap (Table 2). The most dissimilar patterns of kelp and non-kelp distributions were with the unsupervised classification method applied to both types of imagery, with spatial overlaps of 57% and 36%, respectively (Table 2).

3.3. Accuracy of classifications

Unsupervised classification applied to satellite and aerial imageries yielded the lowest overall accuracy (respectively 70% and 66%) and kappa coefficient (respectively 0.261 and 0.273) among the six classification methods used (Table 3). Supervised and visual classifications outperformed unsupervised classifications, as shown by gains of 14% to 24% in overall accuracy (Table 3). Visual classification of aerial imagery was the most accurate method (overall accuracy: 90%; kappa coefficient: 0.757), followed closely by supervised classification of satellite imagery (overall accuracy: 89%; kappa coefficient: 0.744; Table 3).

Visual classification of aerial imagery and supervised classification of satellite imagery exhibited the highest producer's accuracy for the kelp (97%) and non-kelp (95%) classes, respectively, as well as the highest user's accuracy for the non-kelp (91%) and kelp (98%) classes, respectively (Table 3). Producer's accuracy was lowest with unsupervised classification of kelp on aerial imagery (68%) and of non-kelp on satellite imagery (44%) (Table 3). User's accuracy was lowest with unsupervised classification of kelp on satellite imagery (76%) and of non-kelp on aerial imagery (46%; Table 3).

4. Discussion

Spaceborne or airborne infrared imagery is increasingly used to map floating canopies of kelp forests at the sea surface (Bell et al., 2015b; Cavanaugh et al., 2010; Nijland et al., 2019; Stekoll et al., 2007). Canopy maps are then used to estimate subsurface kelp distribution and biomass as well as the influences of various physicochemical factors on them (Bell et al., 2015a; Cavanaugh et al., 2010, 2011) (Friedlander et al., 2007). Completely submerged kelp beds are more difficult to detect because infrared wavelenghts emitted by the sun filter out completely by the time they reach kelp near the seabed. Shorter wavelenghts that better penetrate water have been used to map submerged kelp beds (Casal et al., 2011; Hoang et al., 2015; Sagawa et al., 2012; Simms and Dubois, 2001; Vahtmäe et al., 2012). However, progress in this area has been relatively slow with only a handful of studies that

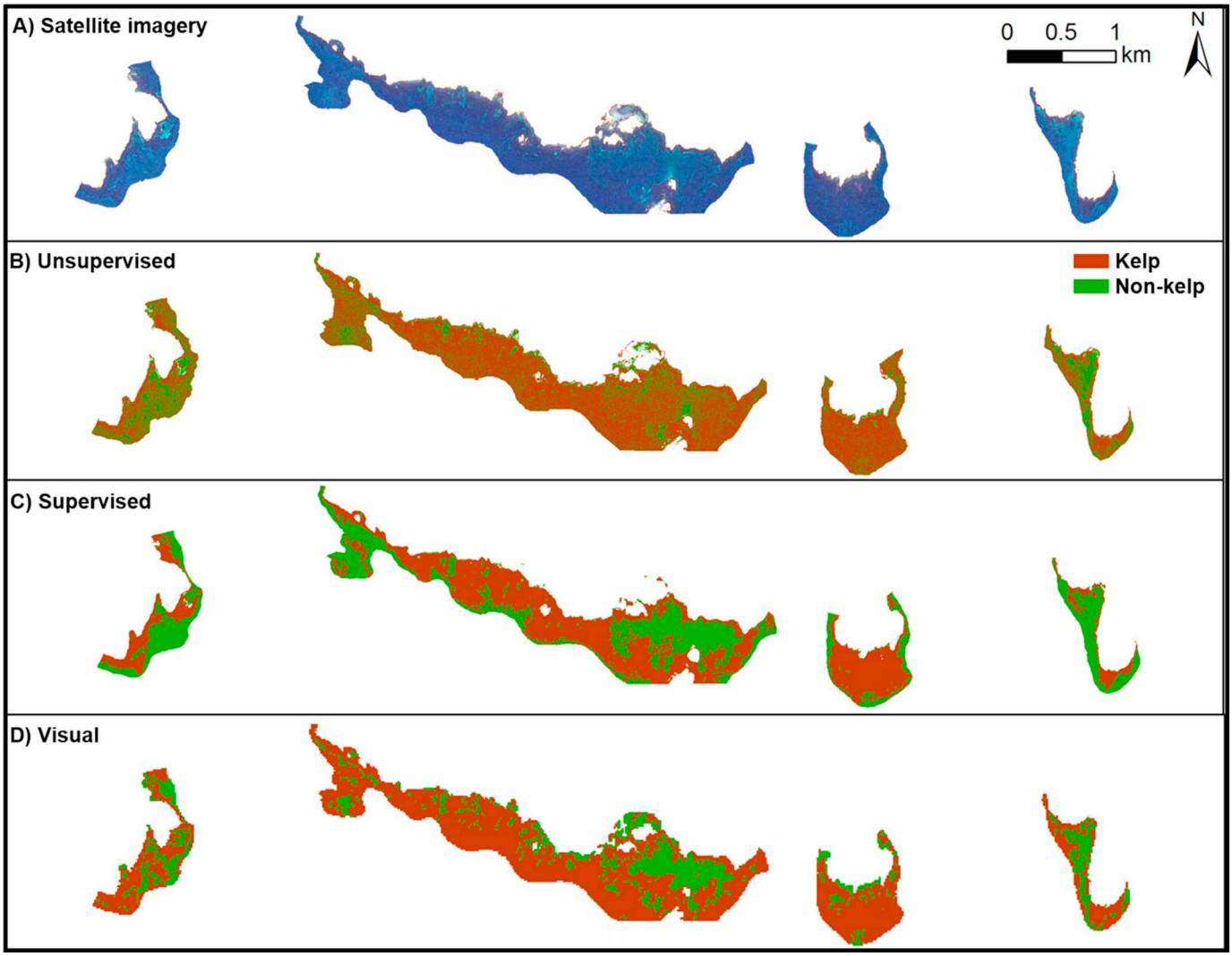

Fig. 5. Satellite (SPOT 7) imagery of the study system in the Mingan Archipelago (from west to east; île à Firmin, île du Havre, île aux Goélands, and Petite île au Marteau) (A) and associated thematic maps obtained with the three image classification methods tested: unsupervised (B), supervised (C), and visual (D).

have specifically quantified and compared the ability to create accurate maps of submerged kelp bed distribution with simple remote sensing and GIS tools (Casal et al., 2011; Gagnon et al., 2008; Uhl et al., 2016). By comparing several accessible, yet simple and robust methods, the present study helps guide map users (sensu Andréfouët, 2008) in the choice of suitable approaches to map completely submerged kelp beds in optically and bathymetrically shallow benthic systems.

The present study demonstrates that satellite and aerial optical imagery, together with basic image classification methods, can produce accurate maps of submerged kelp bed over large (several km^2) tracts of seabed. Overall accuracies of maps from supervised classification of satellite imagery (89%), and visual classification of aerial imagery (90%) matched those of studies of submerged aquatic vegetation in temperate and tropical waters (Gagnon et al., 2008; Green et al., 2000; Mumby and Edwards, 2002; Pasqualini et al., 2005; Valta-Hulkkonen et al., 2003). These accuracies were also above the 85% threshold judged satisfactory for management purposes (Anderson, 1976; Congalton and Green, 2008; Hayes and Sader, 2001). Although in principle both methods can be used interchangeably, some logistical considerations, discussed below, might favor the choice of one method over the other.

Image classification based on unsupervised clustering methods only require minimal user input (typically assigning class names to spectral clusters output by software) and field reference data (Thomson, 1998). Unsupervised methods are particularly accurate when spectral differences among classes are well defined (Rozenstein and Karnieli, 2011; Thomson, 1998). However, this is rarely the case in aquatic habitats since light attenuation by water reduces spectral separability (Green et al., 2000). The application of water column correction techniques can increase the accuracy of unsupervised and supervised classifications (Hoang et al., 2015; Mumby et al., 1998; Sagawa et al., 2012; Zoffoli et al., 2014). However, such techniques are generally too complex to be easily implemented by map users. Unsupervised methods often yield accurate classifications for lakes, wetlands, and seagrass meadows (Dogan et al., 2009; Gullström et al., 2006; Luo et al., 2016), but are generally outperformed by supervised methods in other marine habitats (Calvert et al., 2015; Gagnon et al., 2008; McCarthy and Halls, 2014). In the present study, unsupervised classification of either type of imagery showed poor discrimination between kelp and non-kelp benthic classes, indicating that this method is inadequate to detect submerged kelp beds in the Mingan Archipelago. This method should therefore be used with caution to study kelp bed distribution in systems with similar biological makeup and oceanic conditions.

Supervised classification methods typically produce readily interpretable thematic maps of only the cover classes of interest, and usually outperform unsupervised classification when the training dataset encompasses the spectral variation in each cover classes (Calvert et al., 2015; Green et al., 2000; Lillesand et al., 2014). Supervised techniques perform less well when the spectral signature of one or more classes is highly variable or classes show little spectral differences (Green et al.,

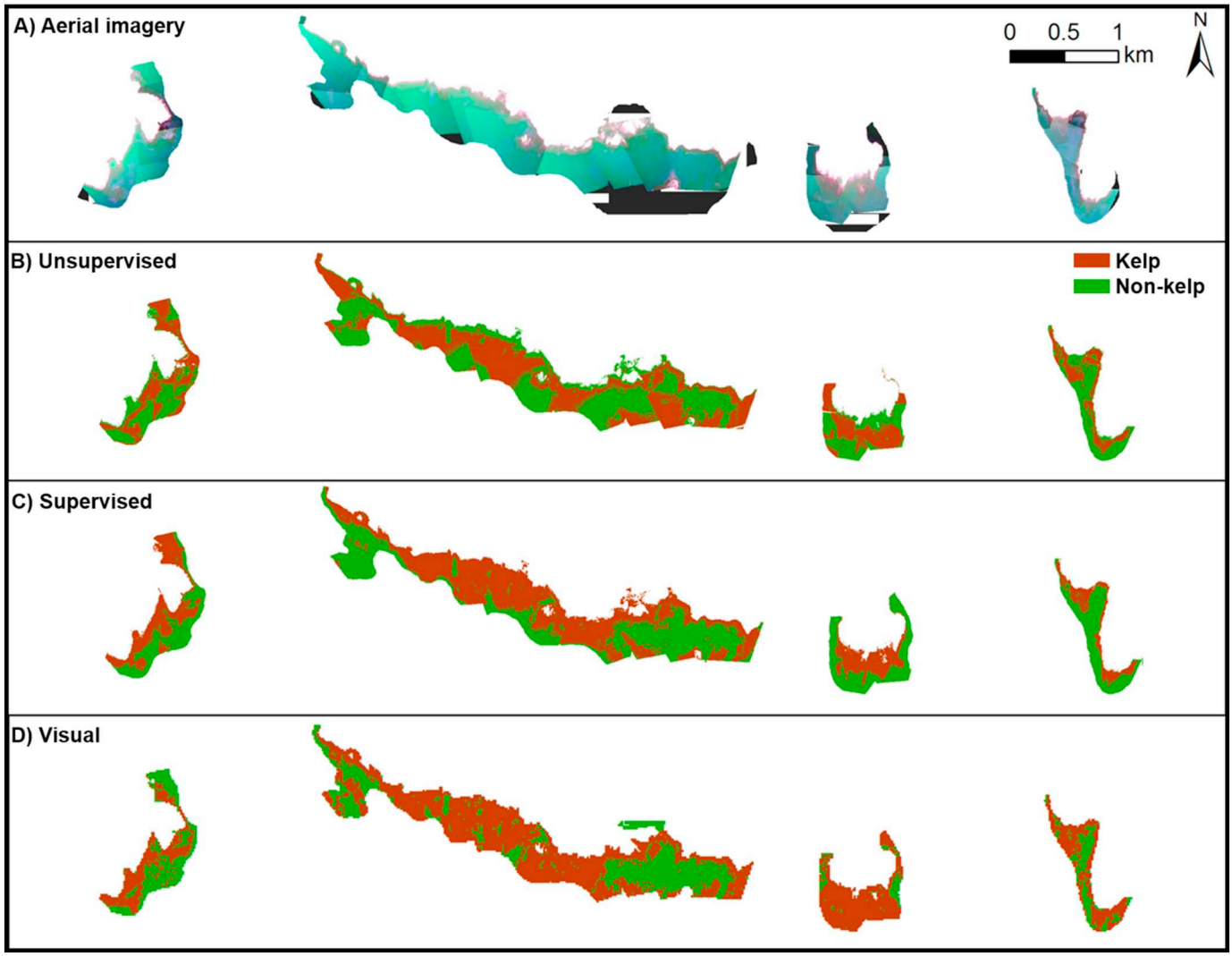

Fig. 6. Aerial imagery (mosaic of aerial photographs) of the study system in the Mingan Archipelago (from west to east; île à Firmin, île du Havre, île aux Goélands, and Petite île au Marteau) (A) and associated thematic maps obtained with the three image classification methods tested: unsupervised (B), supervised (C), and visual (D).

Table 1
Depth-specific proportions of the study area classified as kelp or non-kelp by the six classification methods used (unsupervised, supervised, and visual applied to both types of imagery) or classified differently by at least one classification method.

	Cover		
Depth	Kelp	Non-kelp	Different
0–1 m	30.6%	0.8%	68.6%
1–3 m	43.7%	7.0%	49.3%
3–5 m	22.2%	9.0%	68.8%
5–7 m	6.3%	7.5%	86.2%
Overall	25.3%	7.2%	67.5%

2000; Lillesand et al., 2014; Lu and Weng, 2007). Thus, supervised methods may require more field sampling to ensure that reference data capture variability in the signature of each target class. For the Mingan Archipelago, where kelp and non-kelp benthic classes are spectrally quite distinct, supervised classification performed well, especially with the satellite imagery. Supervised classification of satellite imagery was only ~2% less accurate than visual classification of aerial imagery, the most accurate method (90%). These results are consistent with those of Casal et al. (2011) who mapped kelp beds along the Galician coast with accuracies of 80 to 90% with visual or supervised classifications of satellite imagery (SPOT 4).

Visual classification is somewhat subjective because it centers on the ability of an observer to visually separate benthic classes. Yet, this method has proved highly successful to identify benthic features in different types of subtidal systems worldwide (Andrew and O'Neill, 2000; Drake, 1996; Kendrick et al., 2000; Walker, 2009). As with contextual editing and object-oriented classification (Baatz et al., 2008; Green et al., 2000; Whiteside and Ahmad, 2005), visual classification has the advantage of accounting simultaneously for patch shape, size, and contrast. By relying on the detection of changes in color, in the present case sharp contrasts at kelp bed edges, this method is more robust to cross-image changes in light intensity caused, for example, by variation in water turbidity. In the present study, visual identification of kelp bed edges was more difficult on satellite than aerial imagery because of poorer contrasts on the former and coarseness of the original satellite imagery (6-m pixels color imagery, pansharpened to 1.5-m pixels, and resampled to 1-m pixel). Moreover, kelp bed edges identified with visual classification appeared artificially jagged for both types of imagery, mainly as a result of the minimum mapping unit (MMU; Saura, 2002) of 225 m^2 in the classification grid used with this method. Kelp bed edges were much smoother on the maps created with the unsupervised and supervised classifications, which were based on 1-m-

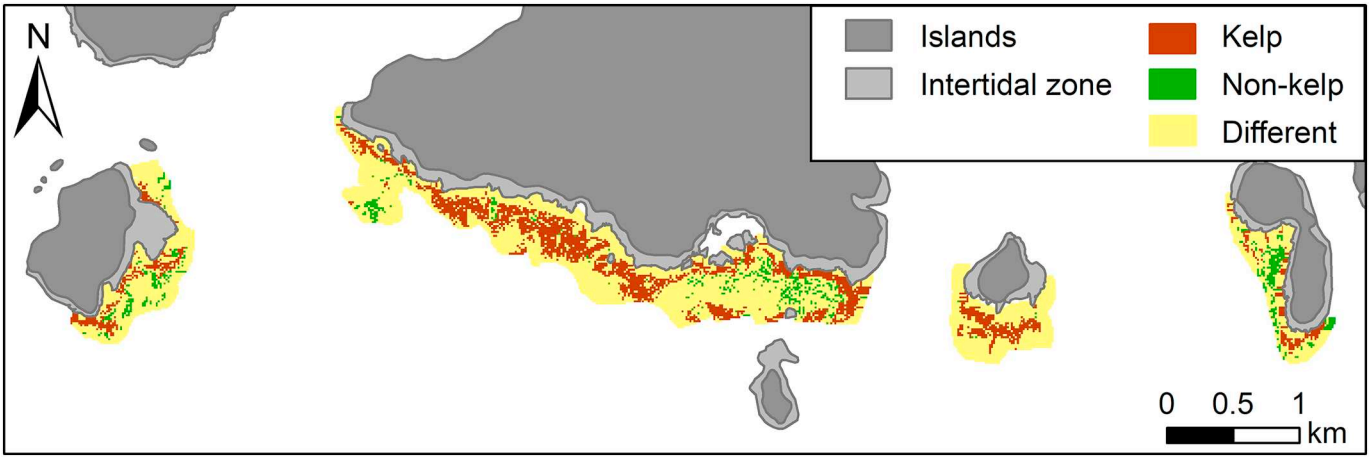

Fig. 7. Location and extent of seabed consistently classified as kelp or non-kelp by the six classification methods used (unsupervised, supervised, and visual applied to both types of imagery) or classified differently by at least one classification method.

pixel images and thus had a much smaller MMU. The size of the grid in the visual classification was intentionally large, to strike an acceptable balance between the minimal spatial scale over which to assess kelp presence (or absence) and the amount of time required to classify the imagery. Yet, because the MMU determines the amount of details within a map, the user's goals and analysis of the information mapped should guide the choice of appropriate MMU and classification method (Saura, 2002). Both the spatial scale of kelp assessment and classification time can be adjusted based on needs and resources. Visual classification necessitated 30 to 45 h of work for each type of imagery (including observer training), while unsupervised and supervised classifications were generally completed within 6 to 10 h.

In the present study, supervised and unsupervised classifications yielded more accurate maps of kelp distribution with satellite than aerial imagery. Aerial imagery presented better contrasts and more contextual details than the satellite imagery, which generally yields higher accuracy in automated classification (Green et al., 2000). However, in the present case variation in contrast and color among the aerial photographs could not be completely corrected for during mosaicking, introducing noise picked up by the unsupervised and supervised classifications. Conversely, satellite imagery showed less pronounced but more uniform contrasts across the study area, making it more suitable than aerial imagery for supervised and unsupervised classification.

Ground truth data is essential to define spectral classes for supervised classification, train the observer for visual classification, and assess image classification accuracy (Green et al., 2000; Lillesand et al., 2014). In the present study, four people completed boat-based ground truthing in about 100 h, including time to familiarize with the sampling gear. This approach enabled sampling a relatively large tract of seabed (~250 ha) much quicker than if scuba-based mapping techniques had been used (Frey and Gagnon, 2015; Gagnon et al., 2004; Lauzon-Guay, 2007; Rinde et al., 2014; Scheibling et al., 1999). Urchin fronts in eastern Canada, including in the Mingan Archipelago, can destroy kelp beds at a rate as high as 4 m mo^{-1} during summer (Gagnon et al., 2004; Lauzon-Guay, 2007; Scheibling et al., 1999). These fast changes in community states stress the necessity of synchronizing ground truthing and image acquisition not to introduce extraneous data variability, an aspect largely disregarded in previous studies (Anderson et al., 2007; Simms and Dubois, 2001). Ground truth data in the present study were collected in mid-July, whereas above sea-level imagery was acquired in early July (aerial) and early August (satellite), respectively. Given the small temporal window (~1 mo) over which all data were acquired, kelp loss to urchin grazing likely contributed very little to the errors of classification reported.

The study area was limited to shallow (< 7 m deep) seabed because light scattering and attenuation by the water column prevented visual separation of kelp and non-kelp benthic classes in deeper water (Green et al., 2000). For the same reason, Simms and Dubois (2001) also limited their assessment of kelp biomass from satellite imagery of the Gulf of St. Lawrence to < 7 m deep. Echo sounders or automated underwater vehicles (AUVs) equipped with cameras could help map macrophyte distribution on deeper seabed (Bewley et al., 2012; Calvert et al., 2015; Marzinelli et al., 2015; Minami et al., 2010). Spectral resolution of both satellite and aerial imagery was insufficient to separate kelp species within beds or kelp beds from monospecific stands of *Desmarestia vidiris*. Thus, stands of *D. viridis* were likely classified as kelp, potentially overestimating kelp cover. However, *D. viridis*

Table 2
Coincidence matrix showing the proportion (%) of overlap in seabed classified as kelp or non-kelp (kelp; *non-kelp*) within each pair of comparison of classification methods used. Bolded values in the bottom row indicate overall proportions of the seabed classified as kelp or non-kelp with each classification method.

	Satellite			Aerial		
	Unsupervised	Supervised	Visual	Unsupervised	Supervised	Visual
Satellite						
Unsupervised	–	–	–	–	–	–
Supervised	68.1; *45.9*	–	–	–	–	–
Visual	78.9; *46.8*	73.0; *52.5*	–	–	–	–
Aerial						
Unsupervised	56.6; *35.9*	63.5; *55.7*	58.3; *36.2*	–	–	–
Supervised	58.9; *37.8*	77.2; *79.6*	62.6; *41.2*	70.0; *66.7*	–	–
Visual	73.5; *38.8*	81.3; *70.6*	80.4; *58.3*	64.0; *50.9*	73.8; *63.2*	–
Overall	**70.8; *29.2***	**58.3; *41.7***	**73.5; *26.5***	**52.0; *48.0***	**53.3; *46.7***	**65.5; *34.5***

Table 3
Classification methods used for mapping kelp and non-kelp benthic classes and associated measures of accuracy. Producer's, user's, and overall accuracies are in %. Numbers in parentheses are the 95% confidence intervals.

	Producer's accuracy		User's accuracy			
	Kelp	Non-kelp	Kelp	Non-kelp	Overall accuracy	Kappa coefficient
Satellite						
Unsupervised	81.1 (± 8.6)	43.9 (± 16.4)	76.0 (± 9.1)	51.4 (± 18.0)	69.5 (± 8.3)	0.261
Supervised	85.6 (± 7.8)	95.0 (± 8.0)	97.5 (± 4.1)	74.5 (± 12.9)	88.5 (± 5.9)	0.744
Visual	92.2 (± 6.1)	53.7 (± 16.5)	81.4 (± 8.0)	75.9 (± 17.3)	80.1 (± 7.2)	0.499
Aerial						
Unsupervised	67.5 (± 10.3)	62.5 (± 16.3)	80.0 (± 9.8)	46.3 (± 14.2)	65.8 (± 8.6)	0.273
Supervised	80.0 (± 8.8)	82.5 (± 13.0)	91.1 (± 6.9)	64.7 (± 14.1)	80.8 (± 7.2)	0.581
Visual	96.6 (± 4.3)	75.6 (± 14.4)	89.6 (± 6.6)	91.2 (± 11.0)	90.0 (± 5.5)	0.757

occurred in < 10% of the 329 benthic photo quadrats acquired around the four islands and was observed without kelp only in 2% of the benthic photo quadrats. These results suggest that a negligible portion of the imagery classified as kelp was actually *D. virdis*. Non-kelp macroalgae, including *D. viridis* and turf algae, were rare within the study area, covering altogether only ~6% of the seabed. Furthermore, 26% of the photo quadrats contained strictly non-kelp macroalgae, and the latter covered only ~17% of the photo quadrats' area. Hence, these quadrats contained mainly bare rock, which the image classification methods used identified as non-kelp. The low abundance of non-kelp macroalgae and predominantly small (a few 10s of cm^2) clusters they formed, suggest that their misclassification as kelp was negligible. More advanced sensors, such as hyperspectral imagers with a higher spectral resolution, could increase the ability to identify and map benthic vegetation, yet would necessitate a more specialized workforce, software, and hardware (Gagnon et al., 2008; Lathrop et al., 2006; Uhl et al., 2016; Valle et al., 2015). The two benthic classes used were sufficient to map the relatively simple and largely dichotomous structure of the benthic system studied. A greater number of classes may be needed to create accurate maps of kelp distribution in systems with more complex benthic structures (e.g. Casal et al., 2011), in which case images with a better spectral resolution than that in the present study (including a greater coverage of the blue to yellow wavelengths) could be required (Green et al., 2000; Mumby and Edwards, 2002).

In the present study, the acquisition of satellite and aerial imagery was planned and executed with the goal of acquiring imagery at times of optimal sea state and water clarity to maximize image quality and the accuracy of the classification maps produced from it (Green et al., 2000, 2017). Therefore, all images were acquired at low tide on cloudless days (to maximize light penetration to, and reflection from, the seabed), prior to zenith (to minimize glare on the imagery), and following several days with low winds and no precipitation (to limit water turbulence and freshwater runoffs). Satellite and aerial images were each acquired on a single day, with less than a month separating both days. These precautions provided the best images possible for our study system, while enabling trustable comparisons between any set of maps. Our results suggest that the classification methods tested could be used to study temporal changes in the distribution and abundance of submerged kelp, provided that consistently high quality can be achieved across an image time series, which, as mentioned above, is best achieved under optimal sea conditions. For studies of change over time, collection of ground-truth data at the time of each image acquisition would be an asset to increase confidence in the information extracted from map products (Green et al., 2000; Lillesand et al., 2014). Preliminary tests with imagery acquired across a range of sea state and turbidity may be required to identify the minimal environmental requirements for accurate classification.

Scuba- and boat-based sampling methods are used extensively for benthic habitat mapping and monitoring (Filbee-Dexter et al., 2016; Himmelman, 1991; Lauzon-Guay et al., 2009; Matarrese et al., 2004; Polovina et al., 1995; Smale and Moore, 2017; Van Rein et al., 2009; Vásquez et al., 2007). These time-consuming and labor-intensive methods are useful to study small-scale (i.e. a few 100s of m^2 over < 5 y) ecological patterns, but present two major limitations for larger-scale applications. First, the methods used for small-scale benthic monitoring often are inadequate to work at large extents because of practical limitations. Second, factors and processes that drive species distribution often differ across scales, and thus results from small-scale studies can rarely be extrapolated to larger scales with confidence (Edwards, 2004; Levin, 1992; Schneider, 2001; Turner et al., 2001). Yet, the development of reliable models of benthic community and ecosystem dynamics is largely based on repeated quantification of species distribution over increasingly large spatiotemporal scales (Lecours et al., 2015; Wiens, 1989). Multiscale monitoring is particularly important for systems that exhibit multiple stable states and for management purposes (Groffman et al., 2006; Moffett et al., 2015; Petraitis and Dudgeon, 2004). Because remote sensing and GIS enable mapping and comparing species distribution over multiple spatial and temporal scales, they are increasingly used to assess the stability and resilience of marine systems (Knudby et al., 2013, 2014; Moffett et al., 2015; Scopélitis et al., 2009).

Kelp-barrens systems often exhibit characteristics of multiple stable state systems, including sudden shifts in, and sharp boundaries between, kelp- and urchin-dominated community states (Filbee-Dexter and Scheibling, 2014; Ling et al., 2015; Moffett et al., 2015). Accurate knowledge of the factors and processes that control the stability and resilience of kelp-barrens systems is crucial to predict and mitigate large-scale shifts of these ecologically and economically important systems (Groffman et al., 2006; Moffett et al., 2015; Petraitis and Dudgeon, 2004). The present study identifies supervised classification of satellite imagery and visual classification of aerial imagery as the top two methods to map kelp distribution, and from there study kelp-barrens dynamics, in the Mingan Archipelago. More broadly, it demonstrates that conventional remote sensing and GIS methods can be used to accurately map and monitor submerged kelp beds over large, yet largely unexplored, spatial and temporal domains. This operational improvement should provide "non-specialist map users" (*sensu* Andréfouët, 2008) like most kelp ecologists, with the ability to address novel questions relating to alternative stable state theory, including the effects of landscape heterogeneity on the stability of community states and basins of attraction (St-Pierre et al., in prep).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jembe.2019.151246.

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